3 The curve C has parametric equations

$$x = \mathrm{e}^t - \tfrac{1}{3}t^3, \quad y = 4\mathrm{e}^{\frac{1}{2}t}(t-2), \qquad \text{for } 0 \leqslant t \leqslant 2.$$

Find, in terms of e, the length of C. [6]